TUTORIAL N°1

Subject: Information System Design **Level:** GLSI2	PRISTINI SCHOOL OF AI	**Academic Year:** 2026/2027 **A.Prof**: Ameni YOUSSFI

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Information System Design

Use Case

Exercise 1:

A company wishes to model, using UML, the training process of its employees in order to computerize certain tasks.

The training process is initiated when the training manager receives a training request from an employee. This employee may optionally consult the catalog of training courses offered by organizations approved by the company. The request is reviewed by the manager, who then sends either an approval or a rejection to the employee.

In case of approval, the manager searches for the appropriate training in the catalogs of accredited courses that he or she maintains.

The employee is informed about the content of the training and is provided with the list of upcoming scheduled sessions. Once the employee specifies their choice, they are registered for the selected session with the corresponding training organization.

If the employee is unable to attend, they must inform the training manager as soon as possible so that the latter can request the cancellation of the registration.

At the end of the training, the employee submits feedback on the course attended as well as a document certifying their attendance. The training manager then verifies the invoice sent by the training organization.

1. **Task required: Draw the appropriate use case diagram.**

Exercise 2:

In view of implementing software dedicated to the textile industry, we are mainly studying some functionalities that allow collecting information on the products developed within the company. All company personnel can consult the system, either to check whether a particular product exists or to freely browse the information.

All consultations must be preceded by a light authentication, in which the person provides their name and department for later statistical purposes. Engineers can perform different update operations on the products they are responsible for: adding, removing, and modifying product information. These operations must be preceded by a more thorough authentication, in which the engineer provides their name, department, and enters a password that is verified by contacting the personnel management system.

All operations (consultations and updates) are recorded in an access log and may optionally be accompanied by printing the accessed documents.

Task: Draw the appropriate Use Case Diagram**.**

Exercise 3:

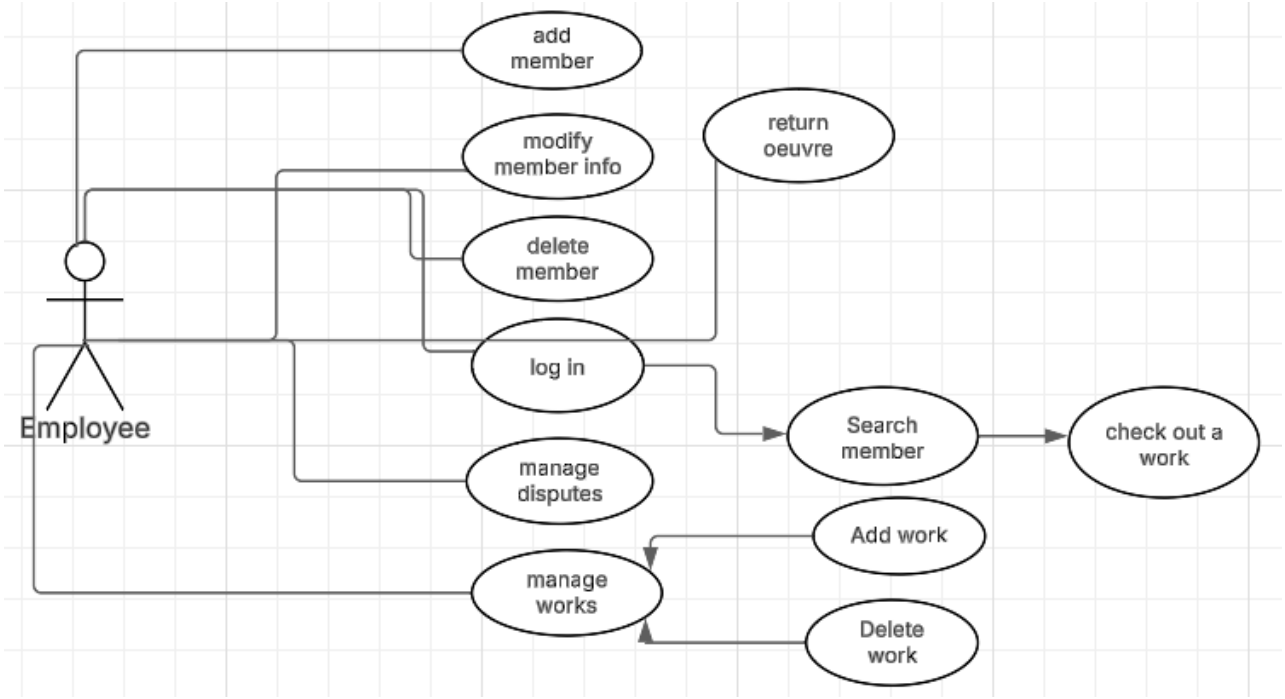
We propose to automate the activities of a small media library that currently has only one employee, although the project manager plans to increase the number of employees in the future.

The media library employee wants the system to allow management of the library's works, members, and loans. The loan of a copy of a given work is limited to three weeks. If the copy is not returned within this period, it generates a dispute. If the copy is still not returned after one-year, legal proceedings are initiated. The employee also wants the future software to allow management of disputes and the

legal proceedings that may arise from them. Access to the system must be protected by a password.

Task :

1. To model the requirements, a student proposed the following diagram:



1. Critique the above Use Case diagram, specifying the pitfalls into which this student has fallen.
2. Propose an alternative solution:

   2.1 Define the actors of the computerized system.

   2.2 Define the appropriate Use Case diagram.